The 2-out-of-3 property: if any two of f, g and g after f are equivalences then so is the third

The Univalent Foundations Program

hott-book:equivalences.tex:local-snapshot-20261011; CC BY-SA 3.0 Unported

Editorial note. The counted statement is the author's 'thm:two-out-of-three' (source lines 612-616, Theorem 4.7.1); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. The retained context is the author's lead-in sentence recalling that equivalences are closed under composition, which is the configuration not spelled out in the two exhibited cases.

Editorial note. The retained counter values (chapter 4, section 7, theorem offset 0) make the counted theorem print as Theorem 4.7.1, matching the original book numbering taken from the label-number dump of the pinned snapshot.

Editorial note. This source directory intentionally contains only the author file, so the retained book macro library is bundled as a per-task copy inside this task directory and referenced from preamble_spans; the copy is verbatim from the same pinned book snapshot and no macro is added, changed or invented. The task directory is used rather than the source directory so that the retained file travels with the card and the author source tree stays untouched.

Editorial note. No retained line refers to a label, citation or figure outside the retained span, so the package declares no external reference.

Editorial scope. One original result of the HoTT Book, counted once: the 2-out-of-3 property for equivalences (Theorem 4.7.1).

Allowed standard prerequisites (editorial).

- Quasi-inverses and equivalences. A function is an equivalence as soon as it has a two-sided homotopy inverse, and conversely.. Both cases in the proof are settled by exhibiting an explicit quasi-inverse.
- Homotopy inverses of equivalences. If e is an equivalence then its inverse opp(e) satisfies the two homotopy identities.. The candidate quasi-inverses are built from opp(g o f) composed with g or f.
- Closure of equivalences under composition. A composite of two equivalences is an equivalence.. It is recalled in the retained lead-in and covers the case in which f and g themselves are the given equivalences.

Source reference. Original theorem 2.4.12: sources/hott-hits/book-labelnumbers-first-edition.tex, source lines 1772–1772. Original result recalled in the retained

lead-in sentence, namely that equivalences are closed under composition. It is a navigation pointer only: the counted theorem's own author proof is retained line for line and nothing is replaced by it. The reference number is taken from the label-number dump of the pinned book snapshot.

We have already seen in original theorem 2.4.12 that equivalences are closed under composition. Furthermore, we have:

Theorem 4.7.1 (The 2-out-of-3 property)**.** *Suppose $f : A \to B$ and $g : B \to C$. If any two of f, g, and $g \circ f$ are equivalences, so is the third.*

Proof. If $g \circ f$ and g are equivalences, then $(g \circ f)^{-1} \circ g$ is a quasi-inverse to f. On the one hand, we have $(g \circ f)^{-1} \circ g \circ f \sim \mathsf{id}_A$, while on the other we have

$$\begin{aligned} f \circ (g \circ f)^{-1} \circ g &\sim g^{-1} \circ g \circ f \circ (g \circ f)^{-1} \circ g \\ &\sim g^{-1} \circ g \\ &\sim \mathsf{id}_B. \end{aligned}$$

Similarly, if $g \circ f$ and f are equivalences, then $f \circ (g \circ f)^{-1}$ is a quasi-inverse to g. □